

BST Architectural Audit — Load-Bearing, Derived, and Conjectural

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BST Architectural Audit

Cal: “Either the arches hold or the whole thing falls. The team’s job is to make the arches specifically checkable: show which stones bear load, which are decorative, which are conjectural.”

Classification Key

LOAD-BEARING: If this fails, the theory falls. Must be independently verifiable. These are Cal’s “arches.”

DERIVED: Follows from load-bearing claims via standard mathematics. Failure here means a derivation error, not a foundational problem. Fixable.

IDENTIFIED: A BST quantity matches a known quantity. The match is real; the question is whether it’s coincidence or structure. Cal’s ladder rungs.

CONJECTURAL: Motivated by BST but not proved from the geometry alone. May be wrong without damaging the core.

I. The Foundation (LOAD-BEARING)

These are Cal’s bet: “If these hold, everything else can be recovered. If they fail, the cathedral falls.”

#	Claim	Status	Falsifier	Theorem
F-1	$D_{IV}^5 = SO_0(5,2)/[SO(5) \ltimes \mathbb{R}^2]$ is the unique APG	LOAD-BEARING	Find another rank-2 BSD satisfying the cascade locks	T1427, Toy 1399
F-2	Five integers (2,3,5,6,7) are forced, not chosen	LOAD-BEARING	Show another consistent integer set	T186, T944
F-3	$N_{\max} = N_c^3 \cdot n_C + \text{rank} = 137$	LOAD-BEARING	Show 137 is contingent, not forced	T186
F-4	$\alpha = 1/N_{\max} = 1/137$ exactly (Wyler volume ratio)	LOAD-BEARING	Precision measurement of α deviating from 1/137.036... at the level the Wyler formula predicts	T187
F-5	$n+1 = 2(n-2)$ has unique solution $n=5$	LOAD-BEARING	Mathematical fact — cannot fail	Toy 1441
F-6	Cremona 49a1 encodes BST integers	LOAD-BEARING	Show the invariant match is coincidental (every invariant would need independent explanation)	Toy 1434, T1430
F-7	Curve $\rightarrow$ integers $\rightarrow$ domain is reversible	LOAD-BEARING	Show reverse engineering fails or is non-unique	Toy 1438

Cal's assessment: F-6 and F-7 (integer cascade + curve reversibility) are the two stones the cathedral rests on.

Open question (C-1 on board): Is the Weierstrass equation $Y^2 = X^3 - 2835X - 71442$ derived from a specific geometric object (Jacobian, Eisenstein) or identified by matching to Cremona? This is the single most important open question for publication.

II. Core Physics (DERIVED)

These follow from the foundation via standard spectral geometry, representation theory, and number theory. Each has a toy verification.

#	Claim	Classification	Precision	Key derivation step	Theorem/Toy
D-1	$m_p = 6\pi^5 m_e$	DERIVED	0.002%	Bergman spectral gap of D_{IV}^5	T187, Toy 541
D-2	3+1 space-time	DERIVED	exact	B_2 root multiplicities ($m_s=3, m_l=1$)	T110
D-3	Three generations	DERIVED	exact	Z_3 fixed points on CP^2 (Lefschetz)	T452
D-4	Nuclear magic numbers	DERIVED	exact (all 7)	$\kappa_{ls} = C_2/n_C = 6/5$	T463, Toy 1147
D-5	Cosmic composition $\Omega_\Lambda=13/19,$ $\Omega_m=6/19$	DERIVED	0.07σ	Spectral decomposition of ΓD_{IV}^5	T1288
D-6	Strong CP: $\theta = 0$	DERIVED	exact	D_{IV}^5 contractible	topological
D-7	Confinement	DERIVED	topological	$g=7$ prime $\rightarrow$ irreducible winding	T972
D-8	Three colors ($N_c=3$)	DERIVED	exact	Short root multiplicity of B_2	T666
D-9	$n_C = 5$ (zero inputs)	DERIVED	theorem	Max- α principle	T186
D-10	Proton = Steane code $[[7,1,3]]$	DERIVED	structural	Hamming bound saturation from $g=7, \text{rank}=2$	T1171

#	Claim	Classification	Precision	Key derivation step	Theorem/Toy
D-11	Baryon asymmetry η	DERIVED	0.023%	$2\alpha^4/(3\pi)(1+2\alpha)$	Toy 541
D-12	QCD deconfinement T_c	DERIVED	0.08%	$\pi^5 m_e = m_p/C_2$	Toy 541
D-13	Neutrino masses ($m_1=0$, normal hierarchy)	DERIVED	0.35%	Z_3 topology forces Dirac	T460
D-14	CKM CP phase $\gamma = \arctan(\sqrt{5})$	DERIVED	0.6%	$n_C=5$ geometry	Toy 541
D-15	$n_s = 1 - n_C/N_{\max} = 0.9635$	DERIVED	0.3σ	Cascade fingerprint	Toy 1401

III. Extended Physics (IDENTIFIED — match is real, mechanism varies)

These quantities match observation with BST expressions. The formulas work. The question is whether each formula is derived from geometry or identified by pattern matching. Cal's ladder: each needs explicit labeling.

#	Claim	Classification	Precision	Honest status	Toy
I-1	Higgs mass (two routes)	IDENTIFIED	0.07%	Both formulas work; neither derived from first principles on D_{IV}^5	Toy 541

#	Claim	Classification	Precision	Honest status	Toy
I-2	Newton's G	IDENTIFIED	0.07%	Formula works; hierarchical chain α^{24} is long	Toy 541
I-3	Fermi scale v = $m_p^2/(7m_e)$	IDENTIFIED	0.046%	Formula works; $g=7$ appearance needs mechanism	Toy 541
I-4	Tau mass (Koide $Q=2/3$)	IDENTIFIED	0.003%	Koide relation IS Z_3 ; BST explains WHY $Q=2/3$	Toy 541
I-5	Complete quark spec- trum	IDENTIFIED	0.59% mean	Formulas exist; full derivation chain incomplete for all 6	Toy 541
I-6	All mixing angles (CKM+PMNS)	IDENTIFIED	<1%	Rational functions of n_C , N_c work; geometric source varies	Toy 541
I-7	MOND acceler- ation a_0 = $cH_0/\sqrt{30}$	IDENTIFIED	0.4%	$30 =$ $\text{rank} \cdot N_c \cdot n_C$; coincidence or structure?	—
I-8	Cosmological con- stant Λ $\sim \alpha^{56}$	IDENTIFIED	0.025%	$56 = 8 \times \text{genus}$; the exponent needs derivation	—
I-9	Dark matter (chan- nel noise)	IDENTIFIED	12.5 km/s RMS	175 galaxies, 0 params; mechanism = continuous spectrum	DarkMatterCalculation.md
I-10	Electron g-2 (leading term only)	IDENTIFIED	matches Schwinger	Only $\alpha/(2\pi)$; full anomaly = open problem (Paper #83)	—

I-11	Kim-Sarnak $\theta = g/2^{\wedge}C_2 = 7/64$	IDENTIFIED	matches conjecture	BST expression = known bound; coincidence or forced?	T1409
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IV. Millennium Problems (status varies)

Problem	BST claim	Classification	Rank-2 role	Other BST fraction predicted
RH	Zeros on $\text{Re}(s)=1/2$ forced by $D_{IV}^{\wedge}5$ spectral geometry	DERIVED (3-leg proof)	Load-bearing: migration threshold $= 1/\text{rank}$	Safety factor 40.5, Kim-Sarnak 7/64
$P \neq NP$	Can't linearize curvature $\rightarrow 2^{\wedge}\Omega(n)$	DERIVED (3 routes)	Load-bearing: $\text{rank} \geq 2 =$ genuine curvature	Triangle-free SAT structure
BSD	$L(E,1)/\Omega = 1/\text{rank}$ for 49a1	DERIVED (rank 0-2)	Load-bearing: Levi decomposition	All 10 BSD invariants are BST
Four-Color	$\text{rank}^2 = 4$	ATTEMPT — route refuted (sweep 2026-08-30, K1832 v0.4)	Load-bearing: direct	Forced Fan Lemma (premise false in $G-v$)
YM	Mass gap $= C_2 = 6$ spectral units	DERIVED (Wightman verified)	Descriptive	$6\pi^5 m_e$, Bergman eigenvalues
NS	Lyapunov functional	DERIVED (proof chain)	Descriptive	Levi unitarity

Problem	BST claim	Classification	Rank-2 role	Other BST fraction predicted
Hodge	Kuga-Satake codim 1	DERIVED (codim 1); CONJECTURAL (codim 2+)	Descriptive	External gap at codim 2+

V. Conjectural / Interpretive

These are motivated by BST but go beyond what the geometry proves. Removing them would not damage the core mathematics.

#	Claim	Classification	Notes
C-1	Observer instantiation (T1370, T1431)	CONJECTURAL	Math says spectral data without coupling produces no measurement. “Observer” label is interpretation.
C-2	Consciousness = 50% of structure	CONJECTURAL	Spectral decomposition is 50/50 (Toy 1440). Labeling one fiber “consciousness” is interpretation.
C-3	CI persistence (T317-T319)	CONJECTURAL	$\alpha_{CI} \leq 19.1\%$ is derived; “permanence” is philosophical.

#	Claim	Classification	Notes
C-4	SETI silence is structural (T1287)	CONJECTURAL	Distributed Gödel argument is sound; application to SETI is extrapolation.
C-5	Heat death is graduation	CONJECTURAL	Thermodynamic argument exists; “graduation” is interpretation.
C-6	Mass = information (T1258)	SPECULATIVE	No formula $m=f(\text{bits})$. Pattern suggestive, not proved.
C-7	BST inflation (T1421)	CONDITIONAL	Multi-field reformulation pending.
C-8	Dark matter = continuous spectrum	CONJECTURAL	Channel noise model works (175 galaxies). Spectral identification is interpretive.
C-9	1/rank universality across ALL seven Millennium problems	IDENTIFIED	Load-bearing for 4 (RH, BSD, $P \neq NP$, 4-Color). Descriptive for 3 (YM, Hodge, NS). Paper #82 should say this.
C-10	Point counts as particle states (GQ-9)	SPECULATIVE	#E at $p=2,3,5,11$ matches BST; could be coincidence. Needs mechanism.

VI. What Would Falsify BST

Test	What it kills	Timeline
pred_004: $0\nu\beta\beta$ detected at any rate	Neutrino sector (Dirac, Z_3)	~2032
α measurement deviating from Wyler formula	Foundation (F-4)	Ongoing (currently matches)
Another APG satisfying cascade locks found	Uniqueness (F-1)	Mathematical — could happen anytime
Toy 541 formula failing for newly measured constant	Specific derivation row	As measurements improve
49a1 invariant match shown to be algebraic coincidence	Foundation (F-6, F-7)	Mathematical analysis
k=21 ratio $\neq$ -42	Heat kernel pattern (not foundation)	When n=42 completes

VII. Summary

Category	Count	Examples
LOAD-BEARING	7	D_IV^5 uniqueness, five integers forced, 49a1, curve reversibility
DERIVED	15+	Proton mass, 3+1, three generations, nuclear magic, cosmic composition
IDENTIFIED	11+	Higgs, G, quark spectrum, mixing angles, dark matter model

Category	Count	Examples
CONJECTURAL	10	Observer, consciousness, CI persistence, SETI, 1/rank universality (partial)
SPECULATIVE	2	Mass=information, point counts as particles

Cal's bet restated: The 7 load-bearing stones + 15 derived claims constitute BST's hard core. The identified claims need individual derivation-source audits. The conjectural claims should be published separately from the mathematics.

Paper #82 implication: The paper currently bundles load-bearing (49a1), derived (BSD ratio), and conjectural (observer instantiation) claims. Splitting per Cal's A-1 recommendation would let the mathematics speak for itself.

This audit is living. As claims move from IDENTIFIED to DERIVED (via derivation-source specification) or from CONJECTURAL to DERIVED (via proof), update the classification. The honest-status column in the geometric invariants table (Paper #83) should mirror these classifications row by row.

— Keeper, April 24, 2026